

UST v5 — Addendum

Theorem T31, Open Questions, and Errata

Niyazi ÖCAL

Independent Researcher, Ankara, Turkey

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1 Theorem T31 — White Hole as Symmetry Center

1.1 Statement

The compact object spectrum (T29) has a natural symmetry center at $N_{\text{eff}} = 0.5$:

White hole	$N_{\text{eff}} = 0.5000$	symmetry center
Star	$N_{\text{eff}} = Cc_b = 0.3665$	Channel C
Neutron star	$N_{\text{eff}} = T_{Om} = 0.2325$	equilibrium (T24)
Black hole	$N_{\text{eff}} = N_b = 0.6335$	Channel Q

(1)

1.2 Symmetry Breaking Origin

At $N_b = Cc_b = 0.5$ (Big Bang pre-state), the tunnel amplitude is minimized:

$$T_{Om}(0.5, 0.5) = e^{-\pi/2} = 0.207880 \quad (2)$$

The universe broke symmetry to $N_b = 0.6335$, leaving the imprint:

$$\Delta T_{Om} = T_{Om}(N_b) - T_{Om}(0.5) = 0.232529 - 0.207880 = 0.024649 \quad (3)$$

1.3 Two-Directional Channel Exchange

Every black hole exchanges information with both channels:

$$Q \rightarrow C \text{ (absorption)} : \text{ amplitude} = T_{Om} = 0.2325 \quad (4)$$

$$C \rightarrow Q \text{ (Hawking)} : \text{ amplitude} = 1 - T_{Om} = 0.7675 \quad (5)$$

The white hole is the object where this exchange is **symmetric**: $T_{Om} = 1 - T_{Om}$, which holds only at $N_b = Cc_b = 0.5$.

1.4 Status

This theorem is **conceptual** — no direct observational confirmation yet. It is consistent with T24 (neutron star compactness) and T29 (compact spectrum).

2 Open Questions — Pending CMB-S4 2028

Three numerical relationships involving $RA = N_b - N_m = 9.38 \times 10^{-6}$ show intriguing proximity to cosmological observables. **These are open questions, not theorems.** The factor of 24 lacks a first-principles derivation.

2.1 Primordial Scalar Power Spectrum (A_s)

$$A_s^{\text{UST}} = 24 \times RA^2 = 24 \times (N_b - N_m)^2 = 2.1116 \times 10^{-9} \quad (6)$$

Value		
UST prediction	2.1116×10^{-9}	
Planck 2018	$2.099 \pm 0.014 \times 10^{-9}$	$\Delta\% = 0.60$

2.2 Tensor-to-Scalar Ratio (r)

$$r^{\text{UST}} = \frac{Cc_b}{24} = \frac{1 - N_b}{24} = 0.015269 \quad (7)$$

Current upper bound: BICEP/Keck $r < 0.036$ — UST prediction is within bounds.

2.3 Primordial Non-Gaussianity (f_{NL})

$$f_{NL}^{\text{local}} = 0.000 \quad (\text{single scalar field}) \quad (8)$$

Planck 2018 constraint: $|f_{NL}^{\text{local}}| < 10$ — consistent.

2.4 Falsification Criteria (2028)

These open questions will be tested by:

- **CMB-S4 / LiteBIRD (2028–2030):** If $A_s \neq 2.111 \times 10^{-9}$ at 5σ , the $24 \times RA^2$ relation is ruled out.
- **Simons Observatory / CMB-S4 (2028):** If $r > 0.020$ or $r < 0.010$ at 5σ , the $Cc_b/24$ prediction is ruled out.
- **SPHEREx / Euclid:** If $|f_{NL}^{\text{local}}| > 1$ at 5σ , the single scalar field assumption is ruled out.

3 Errata to Main Documentation

Location	Error	Correction	Note
Table 2, T5	$\Delta\% = 1.092$	$\Delta\% = 0.10$	Transcription error
Table 2, T29	“exact”	$\Delta\% = 0.185$	Approximate, not exact
Table 2, T6	$\Delta\% = 0.19$	$\Delta\% = 0.75$	Corrected value
Reference 11	Euclid TPDR Shape-Entropy	<i>Remove</i>	Not a published paper

4 Updated Summary Table

Table 1: UST v5 Complete Theorem Status

T#	Content	Predicted	Observed	Status
T29	Compact spectrum $\approx \sqrt{3}$	1.728845	1.732051	$\Delta\% = 0.185$
T30	Solar $\eta = 6\pi N_b C_{C_b}$	4.376223	4.383722	$\Delta\% = 0.171$
T31	White hole $N_{\text{eff}} = 0.5$	0.5000	—	Conceptual
T32a	$\kappa \cdot dt_{\text{UST}} = \pi$	exact	exact	$\Delta\% = 0.000$
T32b	$t_H \times N_b \approx \nu_{Cs}$	9.1920 Gyr	9.1926 GHz	$\Delta\% = 0.007$
$A_s?$	$24 \times RA^2$	2.112×10^{-9}	2.099×10^{-9}	Open, 2028
$r?$	$C_{C_b}/24$	0.0153	< 0.036	Open, 2028
$f_{NL}?$	0 (single field)	0.000	$ f_{NL} < 10$	Open, 2028